

Experiment 8

Problem Statement

You are required to design and implement a CI/CD pipeline to automate application deployment across multiple environments using **Docker** and **Jenkins**. The pipeline must include a **testing stage** to validate the application before deployment.

Perform the following tasks systematically.

Task 1: Application Setup

1. Create a project directory.
 2. Develop a simple web application (index.html) containing:
 - A heading indicating environment (e.g., Dev)
 - A message describing automated deployment
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Task 2: Containerization (Infrastructure as Code)

1. Create a Dockerfile.
 2. Configure it to:
 - Use a web server base image (e.g., Nginx)
 - Copy the application file into the container
 - Expose the required port
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Task 3: Version Control Integration

1. Initialize a repository using **Git**.
 2. Commit the project files.
 3. Push the code to a repository on **GitHub**.
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Task 4: Jenkins Pipeline Configuration

1. Set up a pipeline job in **Jenkins**.
 2. Configure the pipeline to connect to the GitHub repository.
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Task 5: Build Stage

1. Configure the pipeline to:
 - Clone the repository
 - Build a Docker image from the Dockerfile
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Task 6: Testing Stage (Mandatory)

1. Add a testing stage in the pipeline to:
 - Run the application inside a temporary container
 - Validate the application using an HTTP request (e.g., curl)
 - Ensure the pipeline stops if the test fails
 - Remove the test container after execution
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Task 7: Deployment Across Environments

1. Configure deployment stages to:
 - Deploy the application to a Dev environment (port 8080)
 - Deploy the application to a Prod environment (port 9090)
 2. Ensure deployment occurs **only if the testing stage is successful**
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Task 8: Automation Trigger

1. Modify the application content.
 2. Commit and push changes to the repository.
 3. Trigger the pipeline (manually or automatically).
 4. Observe the automated execution of all stages.
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Expected Outcome

- The application should be successfully containerized and deployed.
- The CI/CD pipeline should:
 - Build the application
 - Test it automatically
 - Deploy it only if tests pass
- The application should run consistently in both Dev and Prod environments.

Submission Instructions:

1. Students must upload a single PDF file containing the report of the experiment conducted.
2. The report should contain codes and snapshots of outputs corresponding to each task.
3. The PDF name should be in the form "StudentName_SAPID".